

## **Report:**

# **Grocery Store Management Program**

## **Introduction**

The Grocery Store Management program is designed to simulate a simple system for managing the operations of a grocery store. It provides functionalities for adding and displaying products, managing customers, processing sales, and identifying low-stock products. The program is written in C++ and uses fundamental programming concepts like loops, arrays, and conditional statements to achieve its objectives.

## **Program Features**

### **1. Add Product**

- The program allows the user to add new products to the inventory.
- Each product has the following attributes:
  - Name
  - Category
  - Product ID
  - Stock Availability
  - Price per unit
- **Constraints:**
  - A maximum of 20 products can be added to the inventory.

### **2. Display Products**

- The program provides a functionality to display all the products in the inventory along with their details:
  - Name
  - Category
  - Product ID
  - Stock
  - Price per unit
- This feature enables store management to view the current inventory.

### **3. Add Customer**

- The program allows adding customers to the system with details such as:
  - Name
  - Customer ID
  - Age
  - Contact Information
- **Constraints:**
  - A maximum of 20 customers can be added.

### **4. Display Customer Details**

- This feature displays all the registered customer details:
  - Name
  - Customer ID
  - Age
  - Contact Information

### **5. Sell Products**

- The user can process sales for registered customers by:
  - Inputting the Customer ID to identify the customer.
  - Allowing the customer to select products and specify quantities.
  - Verifying stock availability and updating stock after the purchase.
  - Calculating the total bill for the customer.
- **Constraints:**
  - Each transaction can include a maximum of 5 items.

## **6. Low Stock Products**

- This feature identifies and displays products with stock levels below 5 units.
- It helps store management track items that require replenishment.

## **7. Exit**

- The program ends execution when the user selects this option.

## **Program Flow**

1. The program begins by presenting a menu of options to the user.
2. The user selects an option by entering a number corresponding to the desired functionality.
3. Based on the choice:
  - Products or customers are added to the system.
  - Inventory and customer details are displayed.
  - Sales are processed.
  - Low-stock products are highlighted.
4. The menu continues to loop until the user chooses the "Exit" option.

## **Implementation Details**

- **Data Storage:**
  - Arrays are used to store information about products and customers. The maximum size for each array is set to 20.
- **Control Structures:**
  - do-while loop: Ensures the menu is repeatedly displayed until the user exits.
  - switch statement: Handles user choices and executes corresponding functionalities.
  - for loops: Used to iterate through product and customer arrays for searching, adding, or displaying data.

- **Input Validation:**
  - The program checks for conditions like exceeding the maximum limit of products/customers or stock unavailability during sales.
- **Error Handling:**
  - Messages are displayed for invalid inputs, such as entering a non-existent product ID or customer ID.

## **Conclusion**

The Grocery Store Management program is a useful tool for managing basic store operations. While it is suitable for small-scale implementations, enhancements like dynamic data storage and better error handling can improve its efficiency and scalability. This program serves as a foundation for more advanced store management system.